could possibly have been derived. Looking at the graph, all electronic components have a vertex dual, (with different colors by type), but also the joints have been modeled with vertices on the graph (to cope with the fact that we need to restraint to horizontal and vertical segments for conductive tracks).

While we explain these concepts, we'll also lay the grounds to build an application that takes a graph and displays it nicely on the screen (or, equivalently, on paper).

15.1 Graph Embeddings

Graphs are an amazing data structure: in chapter 14 we have just scratched the surface, talking about *Dijkstra's* and *A**, there are several more cool applications for graphs: you must have heard of knowledge graphs[1], or graph databases[2] like Neo4J, just to name a few that are on hype these days.

But graphs are also used to model more tangible applications, for example a printed circuit board can be represented as a graph, where electronic components are the vertices and conductive tracks on the board (usually made out of copper) are the edges[3].